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TITLE

PFI - PRODUCT FORM OF THE INVERSE

AUTHOR

James H. Christensen

COMPANY

University of Oklahoma
Norman, Oklahoma

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SOURCE LANGUAGE

FOCAL

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PFI - PRODUCT FORM OF THE INVERSE

DECUS Program Library Write-up

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INTRODUCTION

The product form of the Inverse (PFI) provides a means of inverting a matrix with minimal storage and computation requirements. It is based on the fact that, if we already have the inverse B of an $(n \times n)$ matrix A , we can obtain the inverse $\hat{B}$ of a new matrix $\hat{A}$, formed by replacing column r of A by a vector $\underline{x}$, with the formulae:

$$\left. \begin{aligned} \hat{b}_{ij} &= b_{ij} - \frac{y_i}{y_r} b_{rj}, \quad i \neq r \\ \hat{b}_{rj} &= b_{rj} / y_r \end{aligned} \right\} \quad j = 1, \dots, n,$$

where $\underline{y} = B\underline{x}$. Thus, to obtain A^{-1} , we begin by setting B equal to the identity matrix (or any non-singular matrix), and perform the PFI n times, once for each column of A . We note that this procedure requires storage only for the current value of B and the vector $\underline{y}$; the components of $\underline{x}$ are used only to compute a new $\underline{y}$ (in lines 2.2, 6.3 and 6.4). Note also that the subscripting algorithm $B(I, J) = B(I + N * (J - 1))$ requires few actual multiplications to evaluate the current subscript.

A typical encounter is shown, in which the matrix

$$B = \begin{bmatrix} 1 & 0 & -1 \\ .2 & -1 & 0 \\ 0 & .75 & -1 \end{bmatrix}^{-1}$$

is evaluated; note that a negative column number input gives a printout of the current B matrix. The second example obtains the inverse of

$$\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & .05 & -1 \end{bmatrix}$$

by simply altering the first two columns, while the last example inverts

$$\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & .9 & -1 \end{bmatrix}$$

by altering column 2 only. The ability to do this is especially useful when doing parametric studies or linear programming.

C FOCAL V3A

01.01C JAMES H. CHRISTENSEN

01.02C SCHOOL OF CHEM.ENG. & MATLS.SCI.

01.03C UNIV. OF OKLA.

01.04C 202 W. BOYD #23

01.05C NORMAN , OK. 73069

01.10 T '!!', "PROGRAM TO INVERT A MATRIX USING PRODUCT FORM", !

01.20 A "DIMENSION OF MATRIX" N

01.30 F I=1, N; S B(I)=0

01.35 F I=1, N; S B(I+N*I-N)=1

01.40 A "COLUMN" R; I (R) 1.5, 1.5; D 2.0; G 1.4

01.50 T '!', "INVERSE IS:"

01.60 F I=1, N; T !; F J=1, N; T %7.06, " ", B(N*J-N+I)

01.70 T '!', G 1.4

02.10 F J=1, N; S Y(J)=0

02.20 S ISB=-N; F J=1, N; D 6.3

02.30 I (FABS(Y(R))-1.E-10) 6.2; S YR1=1/Y(R)

02.40 S ISB=-N; F J=1, N; S ISB=ISB+N; S BRJ=YR1*B(ISB+R); D 3.0

03.10 F I=1, N; S B(ISB+I)=B(ISB+I)-Y(I)*BRJ

03.20 S B(ISB+R)=BRJ

06.20 T !, "MATRIX IS SINGULAR", '!!'; R

06.30 S ISB=ISB+N; A X; I (X) 6.4, 6.5, 6.4

06.40 F I=1, N; S Y(I)=Y(I)+X*B(ISB+I)

06.50 RETURN

*,

*

PROGRAM TO INVERT A MATRIX USING PRODUCT FORM

DIMENSION OF MATRIX:3

COLUMN:1:1:.2:0

COLUMN:2:0:-1:.75

COLUMN:3:-1:0:-1

COLUMN:-1

INVERSE IS:

1.176471	-0.882353	-1.176471
0.235294	-1.176471	-0.235294
0.176471	-0.882353	-1.176471

COLUMN:1:1:1:0

COLUMN:2:0:-1:.05

COLUMN:-1

INVERSE IS:

1.052632	-0.052632	-1.052632
1.052632	-1.052632	-1.052632
0.052632	-0.052632	-1.052632

COLUMN:2:0:-1:.9

COLUMN:-1

INVERSE IS:

10.00000	-9.000000	-10.00000
10.00000	-10.00000	-10.00000
9.000000	-9.000000	-10.00000

COLUMN: ↑ P

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*W

